FedUPI - Test Report

Federated UPI fraud detection system - run on 20 Oct 2026, 11:48 - Windows, Python 3.12, Node 24

106	106	0	93%
tests run	passed	failed	code coverage

Result: ALL CHECKS PASSED. 106 automated backend tests ran in 23 seconds with 93.4% line coverage of the application code (1284 of 1374 statements). The frontend builds without errors, and all 10 pages and the main interactions were exercised in a real browser session with zero JavaScript errors.

What was tested

Area	Checks	How
Access control	Every protected endpoint refuses requests without a login; health check is public	Automated API tests
Risk policy	Amount tiers at the exact boundaries (69,999 / 70,000 / 99,999 / 100,000) for P2P, merchant and investment; risk % bands; model alone cannot reach High	Automated API tests
Repeat payments	Repeats to the same receiver, bursts, and a friend paid twice, using stored history; history features checked against a second, independent implementation	Automated API + model tests
Uploads and exports	CSV, Excel, JSON, TSV and Word uploads; bad files; export to Excel, Word, PDF, CSV, JSON opened and verified	Automated API tests
Security	Password hashing, token tampering and expiry, SQL-injection strings, duplicate emails, CORS, user data isolation, path-traversal file names	Automated API tests
Model quality	Scoring on 2,100 payments the model never saw; determinism; batch vs single scoring; explanations	Automated model tests
Federated training	Training job from the API, duplicate-start protection, parameter limits, secure aggregation exactness, privacy accountant	Automated tests
Frontend	Production build; 10 pages render; menu, theme toggle, scoring form, export dropdown	Build + live browser run

1. Automated test results by file

Test file	Passed	Failed	Skipped	Time (s)
test_api.py	4	0	0	2.1
test_core.py	7	0	0	3.1
test_e2e.py	67	0	0	16.3
test_model_quality.py	9	0	0	0.8
test_paysim.py	1	0	0	0.0
test_privacy.py	4	0	0	0.0
test_security.py	14	0	0	0.8
Total	**106**	**0**	**0**	**23.2**

Group summary

Test	File	Cases	Passed
test_auth_flow	test_api.py	1	1/1
test_score_endpoint	test_api.py	1	1/1
test_upload_export_roundtrip	test_api.py	1	1/1
test_policy_rules_demo_upload	test_api.py	1	1/1
test_generator_shape_and_banks	test_core.py	1	1/1
test_aggregate_is_weighted_average	test_core.py	1	1/1
test_dp_clip_bounds_update	test_core.py	1	1/1
test_federation_learns	test_core.py	1	1/1
test_metrics	test_core.py	1	1/1
test_policy_tiers_and_risk_percent	test_core.py	1	1/1
test_behavior_features	test_core.py	1	1/1
test_health_is_public	test_e2e.py	1	1/1
test_protected_endpoints_need_login	test_e2e.py	8	8/8
test_amount_tiers	test_e2e.py	24	24/24
test_risk_percent_matches_tier	test_e2e.py	1	1/1
test_model_alone_never_makes_high_below_one_lakh	test_e2e.py	1	1/1
test_invalid_payment_rejected	test_e2e.py	7	7/7
test_repeat_to_same_receiver_is_flagged	test_e2e.py	1	1/1
test_burst_of_payments_is_flagged	test_e2e.py	1	1/1
test_two_payments_to_a_friend_stay_low	test_e2e.py	1	1/1
test_upload_every_supported_format	test_e2e.py	5	5/5
test_bad_uploads_rejected	test_e2e.py	5	5/5
test_unsupported_category_in_file	test_e2e.py	1	1/1
test_path_traversal_filename_is_harmless	test_e2e.py	1	1/1
test_dataset_isolation_between_users	test_e2e.py	1	1/1
test_rescore_replaces_old_results	test_e2e.py	1	1/1
test_export_xlsx_is_readable	test_e2e.py	1	1/1
test_export_docx_is_readable	test_e2e.py	1	1/1
test_export_pdf_csv_json	test_e2e.py	1	1/1
test_export_filters	test_e2e.py	1	1/1

Test	File	Cases	Passed
test_alert_workflow	test_e2e.py	1	1/1
test_simulation_and_overview	test_e2e.py	1	1/1
test_epsilon_endpoint_monotonic	test_e2e.py	1	1/1
test_training_job_runs_and_blocks_duplicates	test_e2e.py	1	1/1
test_generator_is_deterministic_and_in_scope	test_model_quality.py	1	1/1
test_history_features_match_a_second_implementation	test_model_quality.py	1	1/1
test_model_ranks_fraud_above_legitimate_on_unseen_data	test_model_quality.py	1	1/1
test_flagged_recall_and_precision_on_unseen_data	test_model_quality.py	1	1/1
test_policy_invariants_hold_for_every_unseen_payment	test_model_quality.py	1	1/1
test_scoring_is_deterministic_and_batch_equals_single	test_model_quality.py	1	1/1
test_explanations_are_present_for_flagged_payments_only	test_model_quality.py	1	1/1
test_report_matches_active_model	test_model_quality.py	1	1/1
test_tiers_function_edges	test_model_quality.py	1	1/1
test_paysim_mapping	test_paysim.py	1	1/1
test_secagg_sum_exact_and_individual_uploads_masked	test_privacy.py	1	1/1
test_secure_aggregate_matches_plain	test_privacy.py	1	1/1
test_epsilon_properties	test_privacy.py	1	1/1
test_noise_for_target_roundtrip	test_privacy.py	1	1/1
test_password_hash_is_salted_and_not_plaintext	test_security.py	1	1/1
test_tampered_token_rejected	test_security.py	1	1/1
test_expired_token_rejected	test_security.py	1	1/1
test_token_for_deleted_user_rejected	test_security.py	1	1/1
test_sql_injection_in_login	test_security.py	3	3/3
test_registration_rules	test_security.py	1	1/1
test_login_error_does_not_reveal_which_part_is_wrong	test_security.py	1	1/1
test_password_change_flow	test_security.py	1	1/1
test_profile_update_ignores_unknown_fields	test_security.py	1	1/1
test_password_hash_never_returned	test_security.py	1	1/1
test_cors_only_allows_local_frontend	test_security.py	1	1/1
test_token_check_is_timing_safe_function	test_security.py	1	1/1

2. Code coverage

Line coverage measured with coverage.py over the whole `app` package while the full suite ran.

Module	Statements	Missed	Coverage
app/api/auth_routes.py	47	0	100%
app/api/routes.py	225	13	94%
app/config.py	11	0	100%
app/data/behavior.py	39	2	95%
app/data/features.py	20	1	95%
app/data/generator.py	123	0	100%
app/data/paysim.py	38	11	71%
app/main.py	18	2	89%
app/ml/experiments.py	128	28	78%
app/ml/explain.py	7	0	100%
app/ml/federated.py	139	7	95%
app/ml/metrics.py	16	0	100%
app/ml/networks.py	22	0	100%
app/ml/policy.py	22	0	100%
app/ml/privacy.py	19	0	100%
app/ml/secagg.py	26	1	96%
app/services/auth.py	53	2	96%
app/services/datasets.py	50	3	94%
app/services/exports.py	63	1	98%
app/services/scoring.py	75	3	96%
app/services/store.py	183	6	97%
app/services/training.py	50	10	80%
Total	**1374**	**90**	**93.4%**

Lowest-covered modules

- **paysim.py (71%)** - the PaySim loader is only tested with a mapping test on a small fake table; the real 178 MB file is exercised by the training run, not by the suite.
- **main.py (72%)** - the static-file branch that serves the built frontend runs only when a build exists.
- **experiments.py (78%) and training.py (80%)** - the full research run (all baselines) is executed by `train.py`, not by unit tests; the suite covers the quick training path.

3. Model checks on data it never saw

The production model scored `samples/upi_test_transactions.csv`: 2,110 payments, 78 of them fraud, generated with a seed that was never used in training.

Check	Measured	Required by the test
ROC-AUC of the model score	0.979	> 0.95
PR-AUC of the model score	0.700	> 0.50
Recall of moderate + high tiers	64.1%	> 60%
Precision of moderate + high tiers	53.8%	> 35%
Payments >= Rs 1,00,000 that are High	100%	all
Payments < Rs 1,00,000 that are High	0%	none
Payments Rs 70,000-99,999 that are Allowed	0%	none
Allow tier risk % maximum	< 40	< 40
Moderate tier risk % range	40 - 79.9	40 - 79.9
High tier risk % range	80 - 99	80 - 99

Tier distribution on this file: Low 2,017 (28 frauds), Moderate 60 (45 frauds), High 33 (5 frauds).

Other model checks that passed

- The synthetic data generator is deterministic for a given seed, stays in scope (only P2P, merchant and investment, amounts Rs 10 to Rs 2,00,000) and produces 1-8% fraud.
- The repeat-payment features computed by the pipeline match a second, independent implementation on 60 randomly chosen payments.
- Scoring is deterministic; scoring a batch gives the same scores as scoring each payment alone.
- Every flagged payment has at least one reason; no allowed payment carries reasons; model reason weights are between 0 and 1.
- The saved report agrees with the loaded model (same threshold, zero raw rows sent, secure-aggregation error below 1e-4, federated PR-AUC above local-only PR-AUC).
- Secure aggregation reproduces plain averaging, the clipping bound holds, and the privacy accountant behaves correctly (more noise gives smaller epsilon, more rounds larger epsilon).

4. Frontend checks

Check	Result
Production build (Vite)	Passed in about 6 s; JavaScript 639 KB (about 180 KB gzipped), CSS 16 KB
Pages rendered in a live browser session	All 10 pages (Dashboard, Live Monitor, Check Transaction, Upload & Datasets, Alert Queue, Downloads, Federated Lab, Model Insights, Research, Profile) showed their heading and content
Charts	Dashboard 2 charts, Model Insights 9 charts rendered
JavaScript errors / console errors	0 across all pages
Menu	Closed on load; opens on the menu button; closes after choosing a page and with Esc
Theme toggle	Switches light to dark and back; choice is saved
Check Transaction form	Preset Rs 1,20,000 payment returned 81.4% high risk with the amount rule as the first reason
Download menu	Lists Excel, CSV, Word, PDF and JSON
Sign-up from the UI	Verified earlier in the project; sign-in screen reached when no token is stored

Not covered

- No pixel-level visual regression or cross-browser testing was done; the browser pane used here is a single Chromium-based browser.
- The file chooser (drag-and-drop upload) was tested through the API, not by clicking in the browser, because a browser script cannot pick local files.
- No load or stress testing. Single-payment scoring took about 7 ms on a laptop CPU, measured over 100 calls; throughput under many users was not measured.

5. Defects found and fixed during testing

#	Finding	Impact	Status
1	Uploading a Word (.docx) table failed with 'invalid error value specified': the table converter used a pandas option that no longer exists.	Word uploads always rejected (the other four formats were fine).	Fixed in datasets.py; test added and passing
2	Database timestamps used a function deprecated in Python 3.12 (datetime.utcnow), producing about 850 warnings per test run.	No wrong behaviour today; would break on a future Python version.	Fixed (timezone-aware helper); suite now has 1 unrelated warning
3	Test design errors: a huge upload used as a test id overflowed an environment variable; a report threshold was compared with too strict a tolerance.	Test-only problems, no product impact.	Fixed in the tests

Remaining warning

- One library notice remains: Starlette says its test client will move from httpx to httpx2. It affects only the tests and is harmless.

Known limitations (not defects)

- The amount rule makes legitimate payments above Rs 99,999 High risk by design; about 1% of legitimate payments fall in that band and are held.
- QR-phishing, stealth fraud, and the first payment of a repeat burst are detected poorly; this is a model limit measured in the research paper, not a software fault.
- Data are synthetic, so model accuracy figures are not real-world accuracy.

6. How to rerun the tests

From the project folder, in PowerShell:

```
cd Backend
.\.venv\Scripts\python.exe -m pytest -q
```

Optional: run `.\.venv\Scripts\python.exe -m pip install coverage`, then prefix the command with `-m coverage run --source=app` and finish with `-m coverage report` to repeat the coverage measurement. The frontend check is `npm.cmd run build` inside the Frontend folder.

Appendix: every test case

File	Test case	Result	Time (s)
test_api.py	test_auth_flow	PASS	0.23
test_api.py	test_score_endpoint	PASS	0.06
test_api.py	test_upload_export_roundtrip	PASS	1.67
test_api.py	test_policy_rules_demo_upload	PASS	0.12
test_core.py	test_generator_shape_and_banks	PASS	0.22
test_core.py	test_aggregate_is_weighted_average	PASS	0.00
test_core.py	test_dp_clip_bounds_update	PASS	0.00
test_core.py	test_federation_learns	PASS	2.88
test_core.py	test_metrics	PASS	0.01
test_core.py	test_policy_tiers_and_risk_percent	PASS	0.00
test_core.py	test_behavior_features	PASS	0.01
test_e2e.py	test_health_is_public	PASS	0.01
test_e2e.py	test_protected_endpoints_need_login[/api/overview]	PASS	0.01
test_e2e.py	test_protected_endpoints_need_login[/api/transactions]	PASS	0.01
test_e2e.py	test_protected_endpoints_need_login[/api/datasets]	PASS	0.01
test_e2e.py	test_protected_endpoints_need_login[/api/model/report]	PASS	0.01
test_e2e.py	test_protected_endpoints_need_login[/api/federated/status]	PASS	0.01
test_e2e.py	test_protected_endpoints_need_login[/api/papers]	PASS	0.01
test_e2e.py	test_protected_endpoints_need_login[/api/export/flagged?format=csv]	PASS	0.01
test_e2e.py	test_protected_endpoints_need_login[/api/auth/me]	PASS	0.01
test_e2e.py	test_amount_tiers[500-ALLOW-p2p]	PASS	0.10
test_e2e.py	test_amount_tiers[500-ALLOW-merchant]	PASS	0.02
test_e2e.py	test_amount_tiers[500-ALLOW-investment]	PASS	0.02
test_e2e.py	test_amount_tiers[69999-ALLOW-p2p]	PASS	0.02
test_e2e.py	test_amount_tiers[69999-ALLOW-merchant]	PASS	0.05
test_e2e.py	test_amount_tiers[69999-ALLOW-investment]	PASS	0.02
test_e2e.py	test_amount_tiers[70000-STEP_UP_AUTH-p2p]	PASS	0.02
test_e2e.py	test_amount_tiers[70000-STEP_UP_AUTH-merchant]	PASS	0.04
test_e2e.py	test_amount_tiers[70000-STEP_UP_AUTH-investment]	PASS	0.05
test_e2e.py	test_amount_tiers[85000-STEP_UP_AUTH-p2p]	PASS	0.04
test_e2e.py	test_amount_tiers[85000-STEP_UP_AUTH-merchant]	PASS	0.05
test_e2e.py	test_amount_tiers[85000-STEP_UP_AUTH-investment]	PASS	0.04
test_e2e.py	test_amount_tiers[99999-STEP_UP_AUTH-p2p]	PASS	0.04

File	Test case	Result	Time (s)
test_e2e.py	test_amount_tiers[99999-STEP_UP_AUTH-merchant]	PASS	0.05
test_e2e.py	test_amount_tiers[99999-STEP_UP_AUTH-investment]	PASS	0.04
test_e2e.py	test_amount_tiers[100000-BLOCK-p2p]	PASS	0.02
test_e2e.py	test_amount_tiers[100000-BLOCK-merchant]	PASS	0.04
test_e2e.py	test_amount_tiers[100000-BLOCK-investment]	PASS	0.02
test_e2e.py	test_amount_tiers[150000-BLOCK-p2p]	PASS	0.04
test_e2e.py	test_amount_tiers[150000-BLOCK-merchant]	PASS	0.02
test_e2e.py	test_amount_tiers[150000-BLOCK-investment]	PASS	0.04
test_e2e.py	test_amount_tiers[199000-BLOCK-p2p]	PASS	0.05
test_e2e.py	test_amount_tiers[199000-BLOCK-merchant]	PASS	0.04
test_e2e.py	test_amount_tiers[199000-BLOCK-investment]	PASS	0.02
test_e2e.py	test_risk_percent_matches_tier	PASS	0.10
test_e2e.py	test_model_alone_never_makes_high_below_one_lakh	PASS	0.02
test_e2e.py	test_invalid_payment_rejected[bad0]	PASS	0.02
test_e2e.py	test_invalid_payment_rejected[bad1]	PASS	0.03
test_e2e.py	test_invalid_payment_rejected[bad2]	PASS	0.03
test_e2e.py	test_invalid_payment_rejected[bad3]	PASS	0.03
test_e2e.py	test_invalid_payment_rejected[bad4]	PASS	0.04
test_e2e.py	test_invalid_payment_rejected[bad5]	PASS	0.02
test_e2e.py	test_invalid_payment_rejected[bad6]	PASS	0.01
test_e2e.py	test_repeat_to_same_receiver_is_flagged	PASS	0.19
test_e2e.py	test_burst_of_payments_is_flagged	PASS	0.17
test_e2e.py	test_two_payments_to_a_friend_stay_low	PASS	0.06
test_e2e.py	test_upload_every_supported_format[csv]	PASS	0.15
test_e2e.py	test_upload_every_supported_format[xlsx]	PASS	0.18
test_e2e.py	test_upload_every_supported_format[json]	PASS	0.14
test_e2e.py	test_upload_every_supported_format[tsv]	PASS	0.12
test_e2e.py	test_upload_every_supported_format[docx]	PASS	0.26
test_e2e.py	test_bad_uploads_rejected[virus.exe-exe-400]	PASS	0.01
test_e2e.py	test_bad_uploads_rejected[empty.csv-empty-400]	PASS	0.01
test_e2e.py	test_bad_uploads_rejected[bad.xlsx-junk-400]	PASS	0.01
test_e2e.py	test_bad_uploads_rejected[cols.csv-columns-400]	PASS	0.01
test_e2e.py	test_bad_uploads_rejected[big.csv-huge-413]	PASS	0.03
test_e2e.py	test_unsupported_category_in_file	PASS	0.05
test_e2e.py	test_path_traversal_filename_is_harmless	PASS	0.14
test_e2e.py	test_dataset_isolation_between_users	PASS	0.19
test_e2e.py	test_rescore_replaces_old_results	PASS	0.26
test_e2e.py	test_export_xlsx_is_readable	PASS	0.17
test_e2e.py	test_export_docx_is_readable	PASS	0.09
test_e2e.py	test_export_pdf_csv_json	PASS	0.10
test_e2e.py	test_export_filters	PASS	0.10

File	Test case	Result	Time (s)
test_e2e.py	test_alert_workflow	PASS	0.07
test_e2e.py	test_simulation_and_overview	PASS	0.25
test_e2e.py	test_epsilon_endpoint_monotonic	PASS	0.05
test_e2e.py	test_training_job_runs_and_blocks_duplicates	PASS	12.15
test_model_quality.py	test_generator_is_deterministic_and_in_scope	PASS	0.18
test_model_quality.py	test_history_features_match_a_second_implementation	PASS	0.21
test_model_quality.py	test_model_ranks_fraud_above_legitimate_on_unseen_data	PASS	0.10
test_model_quality.py	test_flagged_recall_and_precision_on_unseen_data	PASS	0.00
test_model_quality.py	test_policy_invariants_hold_for_every_unseen_payment	PASS	0.01
test_model_quality.py	test_scoring_is_deterministic_and_batch_equals_single	PASS	0.21
test_model_quality.py	test_explanations_are_present_for_flagged_payments_only	PASS	0.11
test_model_quality.py	test_report_matches_active_model	PASS	0.00
test_model_quality.py	test_tiers_function_edges	PASS	0.00
test_paysim.py	test_paysim_mapping	PASS	0.02
test_privacy.py	test_secagg_sum_exact_and_individual_uploads_masked	PASS	0.00
test_privacy.py	test_secure_aggregate_matches_plain	PASS	0.00
test_privacy.py	test_epsilon_properties	PASS	0.00
test_privacy.py	test_noise_for_target_roundtrip	PASS	0.01
test_security.py	test_password_hash_is_salted_and_not_plaintext	PASS	0.17
test_security.py	test_tampered_token_rejected	PASS	0.07
test_security.py	test_expired_token_rejected	PASS	0.05
test_security.py	test_token_for_deleted_user_rejected	PASS	0.01
test_security.py	test_sql_injection_in_login[' OR '1'='1]	PASS	0.01
test_security.py	test_sql_injection_in_login[admin@example.com' --]	PASS	0.01
test_security.py	test_sql_injection_in_login[x@y.com; DROP TABLE users;--]	PASS	0.01
test_security.py	test_registration_rules	PASS	0.06
test_security.py	test_login_error_does_not_reveal_which_part_is_wrong	PASS	0.10
test_security.py	test_password_change_flow	PASS	0.22
test_security.py	test_profile_update_ignores_unknown_fields	PASS	0.06
test_security.py	test_password_hash_never_returned	PASS	0.05
test_security.py	test_cors_only_allows_local_frontend	PASS	0.01
test_security.py	test_token_check_is_timing_safe_function	PASS	0.01